

Deep Multi-Modal Discriminative and Interpretability Network for Alzheimer's Disease Diagnosis

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Abstract—Multi-modal fusion has become an important data analysis technology in Alzheimer's disease (AD) diagnosis, which is committed to effectively extract and utilize complementary information among different modalities. However, most of the existing fusion methods focus on pursuing common feature representation by transformation, and ignore discriminative structural information among samples. In addition, most fusion methods use high-order feature extraction, such as deep neural network, by which it is difficult to identify biomarkers. In this paper, we propose a novel method named deep multi-modal discriminative and interpretability network (DMDIN), which aligns samples in a discriminative common space and provides a new approach to identify significant brain regions (ROIs) in AD diagnosis. Specifically, we reconstruct each modality with a hierarchical representation through multilayer perceptron (MLP), and take advantage of the shared self-expression coefficients constrained by diagonal blocks to embed the structural information of inter-class and the intra-class. Further, the generalized canonical correlation analysis (GCCA) is adopted as a correlation constraint to generate a discriminative common space, in which samples of the same category gather while samples of different categories stay away. Finally, in order to enhance the interpretability of the deep learning model, we utilize knowledge distillation to reproduce coordinated representations and capture influence of brain regions in AD classification. Experiments show that the proposed method performs better than several state-of-the-art methods in AD diagnosis.

Index Terms—Multi-modal discriminative representation, block-diagonal constraint, generalized canonical correlation analysis, knowledge distillation, Alzheimer's disease.

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I. INTRODUCTION

ALZHEIMER'S disease (AD) is a progressive neurodegenerative disease, which occurs frequently in the elderly [1]. AD patients are manifested as movement disorders, memory decline, language disorders, etc. Since AD is an incurable disease [2], we can only intervene in the early stage of the disease and alleviate the progression of the disease through treatment. Mild cognitive impairment (MCI) is the prodromal phase of AD, mainly associated with cognitive impairment. So it is very critical to identify MCI from person without MCI by early diagnosis. Neuroimaging [3] has been widely used in AD diagnosis, because it can reflect the structural or functional damage of brain in AD and MCI.

In recent years, machine learning methods in the early diagnosis of AD have received more and more attention. Early studies focused on modeling with single modality, such as magnetic resonance imaging (MRI) [4] or positron emission tomography (PET) [5]. However, using single modality is not enough to reveal the complex structure and function change of the brain, while multi-modal data can supplement more potential complementary information from different perspectives. Zhang and Shen et al. [6] utilized combined kernel function with different modalities, including MRI, FDG-PET and cerebrospinal fluid (CSF), to classify patients with AD/MCI. Tong et al. [7] used similarities from MRI, PET, CSF and genes to generate a unified graph for final classification. Subsequently, some work embeds the structural information of multi-modal data into the fusion model. Hao et al. [8] proposed a multi-modal neuroimaging feature selection method with consistent metric constraint based on structural information for AD analysis. Shao et al. [9] proposed a hypergraph based regularization term to model the high-order structure relationship for AD/MCI classification.

There is a common problem that above hierarchical and structural correlation information among different modalities cannot be well exploited by above shallow models due to the high-dimensional non-linear property in neuroimaging [10]. As shown in many recent studies, deep data representation can be more efficient than the shallow architectures in classification by decoupling complex patterns in the inputs [11], [12], and it has advantages in inferring and modeling the hierarchical structure of brain network [13]. Liu et al. [12] presented a novel deep learning method with a zero-masking strategy to fuse modalities for AD diagnosis. Liu et al. [14] constructed

cascaded convolutional neural networks (CNNs) to learn the multi-level and multi-modal features of MRI and PET brain images for AD classification. For complex time series data, deep learning has also shown its effectiveness. Lee et al. [15] developed a multi-modal recurrent neural network that combines cross-sectional neuroimaging biomarkers, longitudinal cerebrospinal fluid and cognitive performance biomarkers to diagnose AD. El-Sappagh et al. [16] proposed an advanced multi-modal multitasking network, which used the patient's time series data to jointly predict AD multi-class progression task and cognitive score regression task. In addition, deep learning has also achieved good performance in solving the problem of incomplete data in multi-modal fusion, especially in missing view reconstruction. Pan et al. [17] imputed missing PET data based on the corresponding MRI data by using 3D Cycle-consistent Generative Adversarial Networks (3D-cGAN) to capture the underlying relationship from data, and diagnosed AD with complete multi-modal data. Gao et al. [18] proposed a task-induced pyramid and attention Gan (TPA-GAN), which integrated pyramid convolution and attention module into GAN for generating the missing PET data. Although deep learning methods have diverse applications in brain disease diagnosis, the structural information embedded in multi-modal data has not been well utilized, which can be regarded as important prior knowledge to select more discriminative features for diagnosis.

Moreover, the interpretability of the diagnosis model has always been a point of concern in the medical field. In neural networks, due to the participation of activation function, the network weight and input cannot be analyzed separately, which leads the clinical interpretability of the deep learning based AD diagnosis model is poor. For example, it is hard to provide the influence of brain regions (ROIs) in AD diagnosis by using the existing deep model. A few studies have attempted to explain the significance of ROI by the prediction model. Liu et al. [12] examined the features acquired from the first hidden layer to identify sensitive ROIs to the AD progression. However, this method ignores the role of the following layers of networks, which is not comprehensive for global analysis.

In this paper, we proposed a novel method named deep multi-modal discriminative and interpretability network (DMDIN) via block-diagonal constraint and knowledge distillation, which consists of two parts: deep multimodal discriminative representation (DMDR) and linear multimodal knowledge distillation (LMKD). The former is designed for accurate diagnosis, while the latter, an approximation of the former, is used to discover biomarkers.

For deep multimodal discriminative representation (DMDR), it projects each modality into a common discriminative space through the class structural information to achieve better performance of AD classification. Firstly, we nonlinearly map the heterogeneous modality to a new representation space through the dedicated MLP. Secondly, we introduce discriminative self-expression in fusion model with block-diagonal constraint. We utilize self-expression learning, in which coefficient matrix is shared, to exploit the similarity of information inherent in the modalities. We further embed structural information in the self-expression to achieve inter-modal and the intra-modal alignment. Thirdly, GCCA is adopted to generate a coordinated representation space,

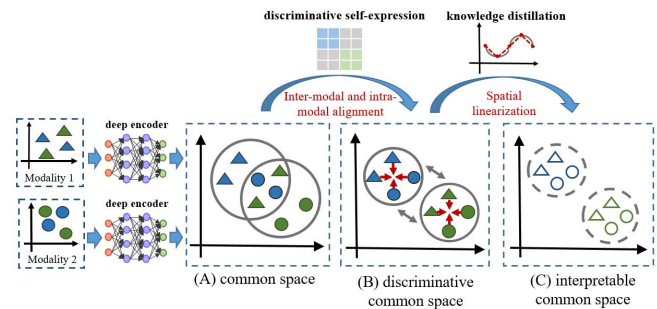


Fig. 1. The discriminative and interpretable common space in the proposed method. (A) the common space with indistinguishable samples. (B) the discriminative common space after discriminative self-expression. (C) the interpretable common space captured by knowledge distillation.

maximizing the correlation coefficients between modalities. The correlation constraint by GCCA and alignment constraint by discriminative self-expression jointly induce MLPs to generate a discriminative common space. It is worth noting that our proposed method is an improved version of the deep generalized canonical correlation analysis (DGCCA) proposed by Benton et al [19]. DGCCA aims to learn nonlinear transformations with multi-view data, which is mainly composed of nonlinear encoders and GCCA constraint. However, due to individual differences and noise in each modality, the sample representation is not uniform, which will further affect the feature combination in generating the common space. Some samples from different classes may be similar in the normal common space generated by DGCCA, as shown in Fig. 1(A). In our method, we introduce discriminative self-expression in fusion model based on DGCCA, which facilitates the mutual representation of intra-class samples and suppresses the mutual representation of inter-class samples to optimize latent space. Compared with the normal common space, the disparity in the distribution of samples of different class trends to be more significant in the discriminative common space, where the intra-class samples are more clustered and inter-class samples are farther away, as shown in Fig. 1(B).

Aiming at exploring the impact of ROIs in AD diagnosis, we reconstruct the above deep network with a linear multi-modal knowledge distillation (LMKD) module. Different from the previous ROIs analysis method that only focuses on partial layer, our method prefers to intuitively establish a linear correlation model between each modality and the coordinated representation, which provides an approximation of the complete model of DMDR. For a certain modality, we use LMKD with single-layer linear mapping to learn the multi-modal representation in the common discriminative space generated from DMDR, as shown in Fig. 1(C). The significant ROIs for the disease diagnosis can be identified by the weights from the LMKD model.

All in all, our method has the following contributions:

- 1) We utilize self-expression learning to reduce the non-uniform in sample representation among different modalities, which promotes more efficient feature fusion in generating common space.
- 2) We develop a discriminative multi-modal fusion framework, which further imposes a block-diagonal constraint on the

deep multi-modal discriminative representation (DMDR) for realizing intra-modal and inter-modal alignment and achieving discriminative common space.

3) We propose the linear multi-modal knowledge distillation (LMKD) module to identify the sensitive biomarkers of AD diagnosis.

4) The experimental results show that our proposed method has promising performance compared with several state-of-the-art methods in AD diagnosis.

The rest of this paper is organized as follows. In Section II, we introduce some related works. Then, we present the proposed model and its optimization in detail in Section III. Datasets, comparison methods and experimental settings are placed in Section IV. In Section V, our experimental results and discussion are displayed. In the end, we draw a summary in Section VI.

II. RELATED WORK

A. Deep Canonical Correlation Analysis

Canonical correlation analysis (CCA) [20] is a widely used fusion method based statistical learning that maximizes the correlation coefficients of two views through linear projection, which can generate structural coordinated spaces for heterogeneous data [21]. Kernel canonical correlation analysis (KCCA) [22] is proposed as an extension of CCA on nonlinear projection, which uses kernel function to project the views into Hilbert space to seek the maximum correlation coefficients. KCCA is limited by fixed kernel, and deep canonical correlation analysis (DCCA) [23] is proposed to maximize the correlation coefficients of two views through deep networks. In DCCA, we define two views as $X_1 \in R^{d_1}$ and $X_2 \in R^{d_2}$, where $d_i, i = 1, 2$ denotes the dimension of the i -th view. The output of the m -layer neural network with c_i units:

$$h_m^{(i)} = g(W_m^{(i)} h_{m-1}^{(i)} + b_m^{(i)}). \quad (1)$$

where $h_1^{(i)} = X_i, i = 1, 2$, $g : R^{c_i} \rightarrow R^{c_i}$ represents the activation function, $W_m^{(i)} \in R^{c_i \times d_i}$ is a matrix of weights, and $b_m^{(i)} \in R^{c_i}$ is a vector of biases. We define the final output of the network as $f_i(X_i)$, and the weights $\theta_i = \{W^{(i)}, b^{(i)}\}$ of these networks are trained through standard back propagation to maximize the CCA objective:

$$(\theta_1^*, \theta_2^*, R_1^*, R_2^*) = \arg \max_{(\theta_1, \theta_2)} \text{corr}(f_1(X_1; \theta_1)R_1, f_2(X_2; \theta_2)R_2). \quad (2)$$

where $R_i, i = 1, 2$ are projection matrix. Further, problem in Eq. (2) is equivalent to:

$$\min_{(R_1, R_2, \theta_1, \theta_2)} \|f_1(X_1; \theta_1)R_1 - f_2(X_2; \theta_2)R_2\|_F^2. \quad (3)$$

B. Self-Expression Learning

Self-expression learning is widely used in subspace clustering [10], [24], which works on latent representations and reflects sample similarity. Ji et al. [24] proposed a stacked auto-encoder based model and adopted self-expression to learn the affinity of data in latent space for clustering. Self-expression can be extended to multi-modal or multi-view problems by seeking a common low-dimensional latent

subspace that captures the most likely consistency among different modalities or views. Gao et al. [10] proposed a framework for cross-model clustering, which incorporates the correlation constraint with a self-expressive layer to make full use of information among the inter-modal data and the intra-modal data. Wang et al. [25] proposed a novel multi-view subspace clustering network, which combines global and local structures with self-expressive layer. According to the self-expression property, a sample can be represented as a linear combination of other samples in the same subspace, we present the equation:

$$ZX = X \quad \text{s.t.} \text{diag}(Z) = 0. \quad (4)$$

where $Z \in R^{n \times n}$ denotes the self-expression matrix, which reflects the similarity between samples. In order to avoid the trivial solution $Z = E$, where E is the identity matrix, we set a constraint $\text{diag}(Z) = 0$.

C. Knowledge Distillation

Knowledge distillation is a model compression and acceleration technology, which transfers the information from a large teacher model into training a small student model without a significant drop in accuracy. According to the form of learning goal, we divide it into three categories: response-based knowledge, feature-based knowledge and relation-based knowledge. For feature-based knowledge, it usually combines deep neural network known as representation learning to construct models. Both the output of the last layer and the output of intermediate layers in deep networks, i.e., feature maps, can be used as the knowledge to supervise the training of the student model [26]. In general, the distillation loss for feature-based knowledge transfer can be formulated as:

$$\text{Loss} = L_F(\phi_t(f_t(X)), \phi_s(f_s(X))). \quad (5)$$

where $f_t(X)$ and $f_s(X)$ respectively denote the representations of the intermediate layers of teacher and student model. And ϕ_t and ϕ_s transform the representations into the same shape. L_F indicates the similarity function.

III. PROPOSED METHOD

The proposed deep multi-modal discriminative and interpretability network (DMDIN) consists of two modules: deep multimodal discriminative representation (DMDR) and linear multimodal knowledge distillation (LMKD). Multi-modal data is reconstructed into discriminative coordinated representation through DMDR, which is composed of three parts: deep encoders, self-expressive layer and GCCA constraint. Although DMDR has a strong discriminative ability, it is still hard to decouple the correlation between features and diagnosis due to its high nonlinearity. Therefore, we further propose LMKD with linear mapping to approximate the DMDR. The proposed framework is shown in Fig. 2.

A. Deep Multi-Modal Discriminative Representation

1) *Deep Encoders*: In recent years, due to the outstanding performance of Transformer, the public attention has gradually returned to MLP [27]. In addition, research shows that an infinite MLP can simulate any neural network. In this paper,

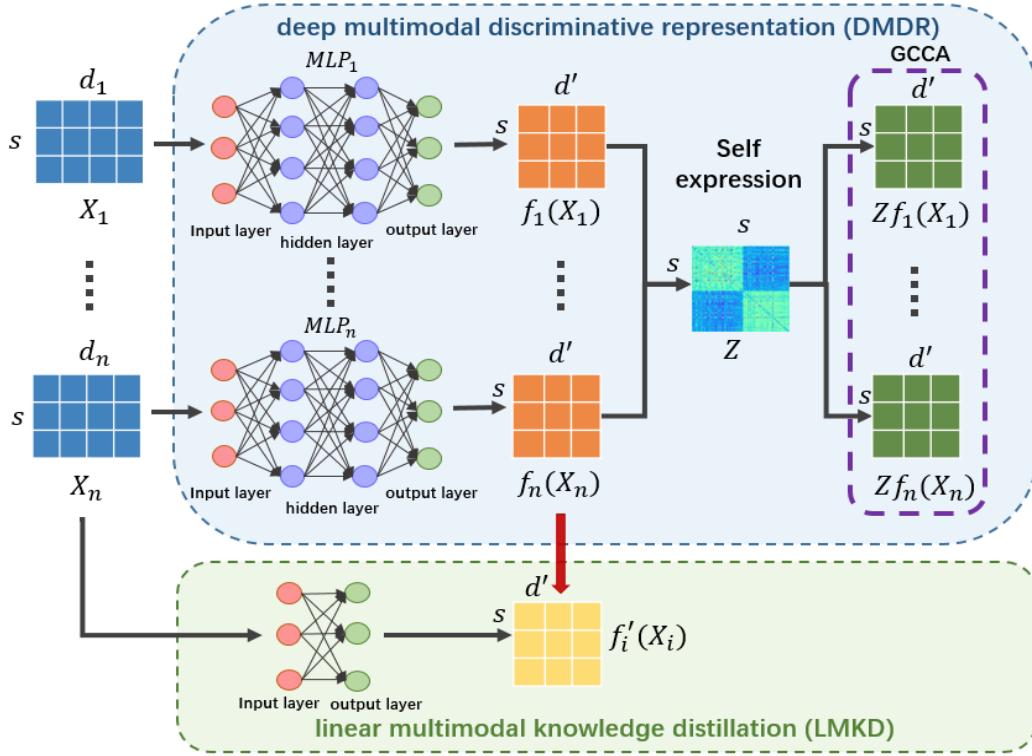


Fig. 2. The framework of the proposed network with DMDR and LMKD. In DMDR, the modality X_i is mapped into the new representation space through the corresponding deep encoder MLP_i and becomes $f_i(X_i)$. The shared self-expression coefficient matrix Z makes $f_i(X_i)$ align samples by introducing discriminative structure information, where Z is added with block-diagonal constraint combined with labels to promote the discrimination. Then, the common space is generated by imposing GCCA constraint on $Zf_i(X_i)$, $i = 1, \dots, n$. In LMKD, the modality X_i learns $f_i(X_i)$ through linear mapping and the result is represented as $f'_i(X_i)$, which can be used to decouple the association between diagnosis label and brain regions.

MLP is used as deep encoder to reconstruct multi-modal data into hierarchical representation. We assume that n modalities with c categories are denoted as $X_i = \{x_i\} \in R^{s \times d_i}$, $i = 1 : n$, where s is the number of samples, and d_i is the corresponding dimension of the i -th modality, while the label can be denoted as $Y \in R^{s \times c}$. Due to the heterogeneity of multi-modal data, we design MLP with different structure for each modality. Specifically, for MLP_i corresponding to X_i , there are five layers of network, in which the number of input layer units is d_i , the hidden layers are $c_i \times d_i$, and the final output layer is d' , $d' \leq d_i$. In addition, ReLU is adopted as the activation function. X_i is nonlinearly mapped into the new representation space according to Eq. (1).

2) Self-Expressive Layer: Due to the self-expression characteristics within the modalities and the similarity of the sample representations between the modalities, we introduce the shared self-expression coefficient matrix $Z \in R^{s \times s}$ in our fusion model, which achieves preliminary sample alignment. We combine the $f_i(X_i)$ with the shared self-expression coefficients, and the Eq. (4) can be rewritten as:

$$Zf_i(X_i) = f_i(X_i) \quad s.t. \text{diag}(Z) = 0. \quad (6)$$

When X_i is input into the model in category order, a diagonal-block structure will appear in the shared self-expression coefficients, where each diagonal block corresponds to one category. With above phenomenon, we utilize the supervision information to embed diagonal-block constraint on the shared self-expression coefficients to further strengthen the intra-modal and inter-modal sample alignment [28]. Considering avoiding the trivial solution and

suppressing the mutual representation among different classes, we define the out-of-class mask matrix as:

$$A = 1_{s \times s} - \begin{bmatrix} 1_{n_1 \times n_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1_{n_c \times n_c} \end{bmatrix} + E_{s \times s}. \quad (7)$$

where $n_i, i = 1 : c$, is the number of samples of class i .

The shared self-expression and its diagonal-block constraint together constitute the loss function of the self-expressive layer:

$$\text{loss}_{self} = \lambda_1 \|Z\|_F^2 + \lambda_2 \sum_{i=1}^n \|Zf_i(X_i) - f_i(X_i)\|_F^2 + \lambda_3 \|A \odot Z\|_F^2. \quad (8)$$

where $\lambda_1, \lambda_2, \lambda_3$ are the regularization parameters respective. $\|\cdot\|_F^2$ denotes the matrix Frobenius norm, $\odot$ indicates the Hadamard product. Specifically, the first term prevents overfitting, the second term guides Z to have the shared self-expression ability, and the third term makes Z discriminative by minimizing its out-of-class self-expression coefficients and the diagonal coefficients.

3) Generalized Canonical Correlation Analysis Constraint: Generalized canonical correlation analysis (GCCA) [29] is an extension of CCA, which aims to map three or more views into a common space to maximize the correlation coefficients between views. In this paper, GCCA is used as the constraint of representation space to generate the common space, which can integrate the potential information in multi-modal data.

The objective function of GCCA as:

$$\min_{(G, Z, \theta_i, R_i)} \sum_{i=1}^n \|G - Zf_i(X_i; \theta_i)R_i\|_F^2 \quad s.t. \quad G^T G = I. \quad (9)$$

where $G = \frac{1}{n} \sum_{i=1}^n Zf_i(X_i; \theta_i)R_i$ is the common space template.

Following the previous GCCA related studies [29], [30], [31], we optimize the objective function of GCCA. We set the i -th modal output as $M_i = Zf_i(X_i; \theta_i) \in R^{s \times d'}$. According to standard notation, we call $M_i^T M_i$ the scatter matrix and $M_i(M_i^T M_i)^{-1}M_i^T$ the projection matrix. Furthermore, we add $r_i I$ to the projection matrix, and it is expressed as:

$$P_i = M_i(M_i^T M_i + r_i I)^{-1}M_i^T. \quad (10)$$

we employ the rank- m SVD [32] on M_i :

$$M_i \xrightarrow{SVD} U_i \Sigma_i V_i^T. \quad (11)$$

where $U_i \in R^{s \times m}$ and $V_i \in R^{m \times d'}$ are respectively left and right singular vectors, and $\Sigma_i \in R^{m \times m}$ is a diagonal matrix composed of the m -largest singular values of M_i . Then the Eq. (10) can be written as:

$$P_i = U_i \Sigma_i^T (\Sigma_i \Sigma_i^T + r_i I)^{-1} \Sigma_i U_i^T = U_i D_i (U_i D_i)^T. \quad (12)$$

where $P_i \in R^{s \times s}$ is symmetric and idempotent, $D_i \in R^{m \times m}$ is a diagonal matrix, which satisfies $D_i D_i^T = \Sigma_i^T (\Sigma_i \Sigma_i^T + r_i I)^{-1} \Sigma_i$. Further, we define the sum of projection matrices as $Q = \sum_{i=1}^n P_i$, and the common space G can be calculated as the left singular vector of rank- m SVD on Q :

$$Q \xrightarrow{SVD} G H L^T. \quad (13)$$

where diagonal matrix $H \in R^{m \times m}$ contains m singular values $k_1, k_2, \dots, k_m$. G and L are the corresponding left and right singular matrices, respectively.

In addition, the objective function Eq. (9) of GCCA can be further expressed as:

$$\min_{(G, Z, \theta_i, R_i)} \sum_{i=1}^n \|G - Zf_i(X_i; \theta_i)R_i\|_F^2 = mn - Tr(G^T Q G). \quad (14)$$

where m is required rank number of SVD, n is number of modalities.

Lemma 1: By considering $R_i = (M_i^T M_i + r_i I)^{-1}M_i^T G$ and the symmetric idempotent matrix Q , the objective function of GCCA finally be optimized as follows:

$$\min_{(G, Z, \theta_i, R_i)} \sum_{i=1}^n \|G - Zf_i(X_i; \theta_i)R_i\|_F^2 \equiv \max_{k_i} \sum_{i=1}^m k_i. \quad (15)$$

Proof:

$$\begin{aligned} \min_{(G, Z, \theta_i, R_i)} \sum_{i=1}^n \|G - Zf_i(X_i; \theta_i)R_i\|_F^2 \\ = \sum_{i=1}^n \|G - M_i(M_i^T M_i + r_i I)^{-1}M_i^T G\|_F^2 \\ = mn - Tr(G^T Q G) \end{aligned} \quad (16)$$

What's more, minimizing the above objective is equivalent to maximizing $Tr(G^T Q G)$. As in Eq. (13), considering the implementation of rank- m SVD on Q , $Tr(G^T Q G)$ can be further optimized to the sum of singular values of Q .

In the end, we train our model by combining the loss function of the self-expressive layer and the optimized GCCA constraint into the final loss function:

$$\begin{aligned} loss_{DMDR} = - \sum_{i=1}^m k_i + \lambda_1 \|Z\|_F^2 + \lambda_2 \sum_{i=1}^n \|Zf_i(X_i) - f_i(X_i)\|_F^2 \\ + \lambda_3 \|A \odot Z\|_F^2. \end{aligned} \quad (17)$$

4) Objective Gradient: The gradient propagation of our model can be roughly divided into two stages, deep network and GCCA mapping. Following [19], we regard the above objective function as a constrained optimization problem, which transforms the optimization objective into a GCCA problem based on the deep network output $M_i = Zf_i(X_i)$. For the network gradients, stochastic gradient descent (SGD) is adopted to update the network weight through back propagation. We focus on calculating the gradient of network output M_i to achieve global optimum. Specifically, the parameters that need to be updated in M_i can be divided into two parts, the multilayer perceptron $f_i(X_i)$ and the self-expression coefficient matrix Z . These two types of parameters are independent, and we calculate the gradient separately.

Lemma 2: For modality i , the network gradients are defined as:

$$\begin{cases} \frac{\partial Loss}{\partial Z} = 2 \sum_{i=1}^n (GR_i^T - Zf_i(X_i)R_iR_i^T)f_i(X_i)^T + 2\lambda_1 Z \\ \quad + 2\lambda_2 \sum_{i=1}^n f_i(X_i)f_i(X_i)^T(Z - I) + 2\lambda_3(A \odot Z)A, \\ \frac{\partial Loss}{\partial f_i(X_i)} = 2(GR_i^T - Zf_i(X_i)R_iR_i^T)Z^T \\ \quad + 2\lambda_2(Z - I)^2 f_i(X_i). \end{cases} \quad (18)$$

Proof: The derivative of Loss with respect to the self-expression coefficient matrix Z is expressed as:

$$\begin{aligned} \frac{\partial Loss}{\partial Z} = \frac{\partial - \sum_{i=1}^m k_i}{\partial Z} + \frac{\partial \lambda_1 \|Z\|_F^2}{\partial Z} \\ + \frac{\partial \lambda_2 \sum_{i=1}^n \|Zf_i(X_i) - f_i(X_i)\|_F^2}{\partial Z} + \frac{\partial \lambda_3 \|A \odot Z\|_F^2}{\partial Z}. \end{aligned} \quad (19)$$

According to the fact $\frac{\partial - \sum_{i=1}^m k_i}{\partial Q} = GG^T$ [33] and the chain rule, the first item can be rewritten:

$$\begin{aligned} \frac{\partial - \sum_{i=1}^m k_i}{\partial Z} &= \frac{\partial - \sum_{i=1}^m k_i}{\partial Q} \frac{\partial Q}{\partial M} \frac{\partial M}{\partial Z} \\ &= 2 \sum_{i=1}^n GG^T (I - P_i) M_i (M_i^T M_i + r_i I)^{-1} f_i(X_i)^T \\ &= 2 \sum_{i=1}^n (GR_i^T - Zf_i(X_i)R_iR_i^T)f_i(X_i)^T \end{aligned} \quad (20)$$

where $R_i = (M_i^T M_i + r_i I)^{-1}M_i^T G$ id defined in Lemma 1.

The remaining three items in Eq. (19) are calculated by F-norm derivation, and the Eq. (19) can be expressed as:

$$\begin{aligned} \frac{\partial Loss}{\partial Z} &= 2(GR_i^T - Zf_i(X_i)R_iR_i^T)f_i(X_i)^T + \lambda_1 \frac{\partial Tr(Z^T Z)}{\partial Z} \\ &\quad + \lambda_2 \frac{\partial Tr((Zf_i(X_i) - f_i(X_i))^T (Zf_i(X_i) - f_i(X_i)))}{\partial Z} \\ &\quad + \lambda_3 \frac{\partial Tr((A \odot Z)^T (A \odot Z))}{\partial A \odot Z} \frac{\partial A \odot Z}{\partial Z} \\ &= 2 \sum_{i=1}^n (GR_i^T - Zf_i(X_i)R_iR_i^T)f_i(X_i)^T + 2\lambda_1 Z \\ &\quad + 2\lambda_2 \sum_{i=1}^n f_i(X_i)f_i(X_i)^T (Z - I) + 2\lambda_3 (A \odot Z)A \end{aligned} \quad (21)$$

Similar to above, the derivative of Loss with respect to each output layer $f_i(X_i)$ can be calculated by the chain rule:

$$\begin{aligned} \frac{\partial Loss}{\partial f_i(X_i)} &= \frac{\partial - \sum_{i=1}^m k_i}{\partial f_i(X_i)} + \frac{\partial \lambda_1 \|Z\|_F^2}{\partial f_i(X_i)} \\ &\quad + \frac{\partial \lambda_2 \sum_{i=1}^n \|Zf_i(X_i) - f_i(X_i)\|_F^2}{\partial f_i(X_i)} + \frac{\partial \lambda_3 \|A \odot Z\|_F^2}{\partial f_i(X_i)} \\ &= \frac{\partial - \sum_{i=1}^m k_i}{\partial M_i} \frac{\partial M_i}{\partial f_i(X_i)} \\ &\quad + \lambda_2 \frac{\partial Tr((Zf_i(X_i) - f_i(X_i))^T (Zf_i(X_i) - f_i(X_i)))}{\partial f_i(X_i)} \\ &= 2(GR_i^T - Zf_i(X_i)R_iR_i^T)Z^T \\ &\quad + 2\lambda_2 (Z - I)^2 f_i(X_i) \end{aligned} \quad (22)$$

B. Linear Multi-Modal Knowledge Distillation Module

In order to improve the interpretability of the model, we propose linear multi-modal knowledge distillation (LMKD) under the inspiration of knowledge distillation [34]. Knowledge distillation can extract compact data pattern from complex models through model simplification. In recent years, many studies have used knowledge distillation to increase the interpretability of complex models [35], [36], [37].

In this paper, each modality has a corresponding LMKD. As increasing deep of Linear MLP does not increase its expressive power [38], LMKD is composed of single-layer linear mapping, which compresses the complex weights. Its structure is consistent with the input layer and output layer of MLP in DMDR. After the DMDR is trained, we fix the parameters of the model. Then, we distill knowledge from the multi-modal representation under DMDR into LMKD through the loss function as follows:

$$Loss_{LMKD_i} = MSE(f_i(X_i), f'_i(X_i)). \quad (23)$$

where $f_i(X_i)$ and $f'_i(X_i)$ are the representations under DMDR and LMKD respectively, and MSE is the mean square error.

LMKD retains the potential relationship learned from multi-modalities, and the feature weights of brain regions can be directly obtained to support ROIs analysis.

Algorithm 1 Deep Multi-Modal Discriminative Representation

Input: Training Set $X_i, i = 1 : n$, Training Label Y , Number of layers

c , Regularized parameter $\lambda_1, \lambda_2, \lambda_3$, Learning rate η .

Output: Train representation $f_i(X_i), i = 1 : n$

Method:

01. Initialize network parameters $Z, \theta_i \sim N(0, 1), i = 1 : n$;
02. Generate out-of-class mask matrix A with label Y by Eq. (7);
03. For epoch i
04. For modality j
05. $M_i \leftarrow$ forward pass X_i with $W = \{\theta_i, Z\}$
06. end
07. $R_1, \dots, R_n, G \leftarrow gcca(M_1, \dots, M_n)$
08. Calculate gradient ∇Z by Eq. (18)
09. $Z' \leftarrow Z - \eta \nabla Z$
10. Calculate gradient $\nabla f_i(X_i)$ by Eq. (18)
11. $\nabla \theta_i \leftarrow backprop(\nabla f_i(X_i), \theta_i)$
12. $\theta'_i \leftarrow \theta_i - \eta \nabla \theta_i$
13. $Z \leftarrow Z', \theta_i \leftarrow \theta'_i$
14. end
15. Return $f_i(X_i), i = 1 : n$;

Algorithm 1 illustrates the DMDR method for training phase. The GCCA constraint and self-expression layer jointly induce MLPs to generate a discriminative common space. As described above, in the training phase, the training set and training labels are used to train the network for re-representation. We concatenate the obtained common subspaces and adopt SVM as the classifier for diagnose. In the testing phase, we make use of the trained network (MLPs) and the trained SVM with test set to predict the stages of AD.

TABLE I

STATISTICAL INFORMATION OF SUBJECTS FROM ADNI-I

Subjects	AD	MCI-C	MCI-NC	NC
Number	51	43	56	52
Gender(M/F)	33/18	28/15	39/17	34/18
Age	75.2	75.8	74.7	75.3
education	14.7	16.1	16.1	15.8

TABLE II

STATISTICAL INFORMATION OF SUBJECTS FROM ADNI-II

Subjects	AD	MCI	NC
Number	160	542	211
Gender(M/F)	95/65	294/247	109/101
Age	75.2	72.5	76.1
education	15.9	16.3	16.4

IV. EXPERIMENTS

A. Datasets

In this paper, the proposed network is validated on two datasets obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) (<http://adni.loni.usc.edu/>). The demographic characteristics of these two datasets can be found in Table I and Table II.

The first dataset is named ADNI-I, which contains 202 subjects, including 51 Alzheimer's disease (AD) subjects, 99 mild cognitive impairment (MCI) subjects and 52 normal control (NC) subjects. In addition, the 99 MCI patients can be further divided into 43 MCI converts (MCI-C) subjects and 56 MCI not-converts (MCI-NC) subjects. The former will convert to AD patients within 18 months, while the latter will remain unchanged. In our work, we select MRI, PET and CSF as three modalities to build our model.

The second dataset is named ADNI-II. It contains 913 subjects, including 160 AD subjects, 542 MCI subjects and 211 NC subjects. We adopt three modalities, i.e. VBM, FDG and AV45.

For the above two datasets, we uniformly adopt the same preprocessing method to process brain images and extract brain regions (ROIs) according to [39]. By using the AAL template, we obtain features of 106 ROIs for the sample in each modality. Then, we remove the cerebellum related brain areas and use the remaining 90 ROIs as features in both datasets for experiments.

B. Experimental Settings and Comparison Methods

In our experiment, we adopt 10-fold cross validation strategy to validate the proposed method and conduct AD diagnosis with different tasks, including AD vs. NC, MCI vs. NC, and MCI-C vs. MCI-NC, to verify the performance of the different models. In addition, we use SVM to classify the fused samples. The parameter r_i in Eq. (10) is selected in the range of $\{1, 1e-1, 1e-2, 1e-3, 1e-4, 1e-5, 1e-6, 1e-7, 1e-8, 1e-9\}$, while the parameters $\lambda_1, \lambda_2, \lambda_3$ are set in the range of $\{1e-4, 1e-3, 1e-2, 1e-1, 0, 1e0, 1e1, 1e2\}$. The grid search is adopted to find the optimal parameter combination. Diagnostic performance is measured by the following metrics: accuracy (ACC), the sensitivity (SEN), the specificity (SPE) and the area under curve (AUC). Our code is available at <https://github.com/xbl-NUAA/DMDIN>.

In order to better evaluate the performance of our proposed method, we select comparison methods in AD diagnosis from two aspects, i.e. multi-modal based methods and deep learning based methods. We choose 17 AD diagnosis methods, including Support Vector Machine (SVM) [40], Multi-kernel Learning (MKL) [6], Visual Geometry Group (VGG) [41], Graph Convolutional Network (GCN) [42], Integrated Network (VGM), Deep Convolutional Auto-encoder (DCAE) [13], Canonical Correlation Analysis (CCA) [20], Deep Canonical Correlation Analysis (DCCA) [43], Self-supervised Graph Neural Networks with Canonical Correlation Analysis (CCA-SSG) [44], Self-paced Multi-Modal Rank Minimization (SPMRM) [39], Joint Multi-Modal Longitudinal Regression and Classification (JMMLRC) [45], Discriminative Block-diagonal Low-rank Representation (BDLRR) [28], Deep Subspace Clustering Networks (DSCN) [24], Cross-Modal Subspace Clustering via Deep Canonical Correlation Analysis (DSCN-CCA) [10], Nonlinear Graph Fusion (NGF) [7], Multi-modal Feature selection with Consistent metric Constraint (MFCC) [8] and Hypergraph based Multi-task Feature Selection (HMFS) [9]. We briefly introduce the above methods and describe the experiment settings in this paper.

1) SVM is a classic classifier, which determines the decision boundary by solving the maximum margin hyperplane of

learning samples. In the experiment, we concatenate the data of three modalities for classification.

2) MKL is an extension of SVM, which projects each modality into a high-dimensional space through different kernel functions and performs linear fusion. SVM is finally used to classify the fused modalities.

3) VGG is a classical deep convolutional network, which adopts a very small convolutional kernel to construct the network. In the experiment, we adopt three VGG16 as the feature extractors of three modalities.

4) GCN is a neural network based on graph structure, which studies brain connectivity through regular Euclidean domains. In this paper, three modalities are as three channels, and the Euclidean distance of nodes is calculated to construct the adjacency matrix. The cross entropy is used as the loss function to predict the label.

5) VGM is an ensemble method that adopts VGG, GCN, and MLP as encoders for each modality and integrates their outputs for classification.

6) DCAE is an auto-encoder network, which models hierarchical structure of brain networks. In this paper, we adopted three DCAE networks to extract and fuse the features of different modalities.

7) CCA is a classical linear multi-modal fusion method, which adopts linear mapping to find a common space that can maximize the correlation coefficients of different views.

8) DCCA makes nonlinear mapping of different modalities through neural network to maximize the correlation coefficients of these modalities.

9) CCA-SSG utilizes graph network for self supervised representation learning, and optimizes feature level objectives combined with canonical correlation analysis.

10) SPMRM adopts multi-modal rank minimization with self-paced learning for revealing the latent correlation across different modalities and fuses the modalities with MKL.

11) JMMLRC performs classification and regression tasks simultaneously by using a variety of regularizations with combined modalities. In this paper, we only consider classification tasks.

12) BDLRR is a semi-supervised representation method, which jointly optimizes the problem of shrinking the unfavorable representation and strengthening the compact block-diagonal representation through block-diagonal constraint.

13) DSCN is an autoencoder based deep network that introduces a self-expressive layer between the encoder and decoder to achieve subspace clustering.

14) DSCN-CCA is a multi-modal extension of DSCN, which introduces canonical correlation analysis in parallel encoders to ensure the similarity of multi-modal subspaces to achieve cross-modal subspace clustering.

15) NGF adopts random forest for pairwise modal similarity and introduces a nonlinear graph to combine the similarity for multi-modal fusion.

16) MFCC measures the similarity of multi-modalities through random forest strategy and considers the group sparsity regularization and the sample similarity to select multi-modal features.

17) HMFS utilizes hypergraph to construct the high-order structure relationship among subjects, and it adopts multi-task

TABLE III
THE CLASSIFICATION RESULTS OF OUR METHOD AND OTHER COMPARED METHODS ON ADNI-I

Method	AD vs. NC				MCI vs. NC				MCI-C vs. MCI-NC			
	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
SVM	87.45	92.17	83.73	94.58	66.79	75.58	55.38	73.90	64.56	60.58	70.96	69.93
MKL	90.27	95.47	88.74	95.97	71.50	80.66	55.31	75.75	67.78	64.02	71.93	70.27
VGG	88.36	92.90	87.57	88.07	76.61	83.02	63.16	76.82	63.78	61.45	66.83	66.27
GCN	93.27	98.00	86.91	93.54	82.72	87.75	71.16	82.22	64.79	55.02	75.67	70.67
VGM	90.45	92.90	87.38	92.33	77.54	85.45	61.92	77.62	63.18	46.02	82.86	63.21
DCAE	90.29	96.34	85.80	96.28	72.32	85.82	47.83	75.32	67.78	59.16	77.86	71.74
CCA	87.36	91.57	81.47	93.78	75.50	81.12	65.26	76.83	69.43	60.16	75.79	69.35
DCCA	91.09	96.57	84.54	94.89	80.47	84.93	71.50	82.49	74.68	62.83	83.61	75.89
CCA-SSG	89.01	92.07	88.05	95.75	85.62	91.11	67.11	80.01	76.21	67.51	80.97	76.79
SPMRM	95.09	95.50	94.33	96.00	82.87	90.12	67.67	81.58	78.84	71.89	85.71	77.18
JMMLRC	90.36	95.47	89.23	96.63	70.21	77.51	56.83	75.31	64.88	56.69	75.38	64.07
BDLRR	92.02	95.58	90.22	94.43	70.33	90.38	43.59	66.96	64.57	55.63	75.55	66.67
DSCN	90.01	96.67	84.14	94.41	73.38	87.09	48.58	73.95	67.12	63.33	73.71	68.64
DSCN-CCA	93.09	95.14	91.23	95.61	76.48	85.78	60.75	78.05	70.98	64.17	78.17	72.82
NGF	89.84	92.57	90.64	95.03	69.63	85.68	42.84	73.48	67.56	52.78	80.42	67.37
MFCC	93.28	95.50	90.67	94.00	81.75	90.12	65.95	81.84	74.56	64.83	83.13	72.45
HMFS	91.36	93.83	89.02	94.21	76.21	84.77	57.53	76.93	73.73	69.16	79.40	70.97
OUR	96.75	98.57	95.50	98.38	86.56	93.34	72.38	83.73	84.38	80.54	86.90	77.31

TABLE IV
THE CLASSIFICATION RESULTS OF OUR METHOD AND OTHER COMPARED METHODS ON ADNI-II

Method	AD vs. NC				MCI vs. NC			
	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
SVM	82.41	88.69	74.15	87.82	64.14	67.89	54.67	67.44
MKL	85.27	90.29	65.15	84.40	74.55	80.50	67.92	81.34
VGG	86.49	92.86	78.12	85.49	62.16	76.19	43.75	59.97
GCN	83.78	89.78	75.00	88.66	60.32	80.68	30.42	60.32
VGM	85.16	88.69	81.30	91.47	64.48	77.26	45.84	66.19
DCAE	86.79	90.99	73.81	90.61	74.29	83.33	44.75	66.32
CCA	84.10	92.68	72.05	90.75	75.04	85.98	42.29	75.25
DCCA	86.48	90.90	80.00	94.16	76.63	83.95	52.79	78.19
CCA-SSG	88.16	95.00	80.57	92.22	80.28	92.73	37.50	70.22
SPMRM	88.02	94.14	80.00	97.21	84.14	94.31	55.26	89.10
JMMLRC	89.22	94.07	82.76	94.78	66.87	81.51	36.69	67.25
BDLRR	88.08	89.52	84.78	86.07	80.23	93.47	47.40	70.36
DSCN	88.13	93.51	81.26	93.82	78.34	83.17	61.23	80.94
DSCN-CCA	88.40	92.10	85.01	93.35	80.62	90.10	47.92	83.38
NGF	84.64	87.14	81.04	90.35	68.78	79.94	40.98	65.10
MFCC	89.48	90.95	88.12	95.05	78.79	85.93	53.44	79.99
HMFS	89.21	91.92	86.18	94.99	78.92	83.58	62.80	76.75
OUR	92.45	94.27	90.44	96.27	86.75	95.15	67.86	91.18

strategy to select multi-modal common features for multi-modal diagnosis.

V. RESULTS AND DISCUSSION

A. Experimental Settings and Comparison Methods

The experimental results of our method and comparison methods on ADNI-I and ADNI-II are shown in Table III, and Table IV, respectively. Specifically, on ADNI-I, our method achieves the ACC values of 96.75%, 86.56% and 84.38% on the tasks of AD vs. NC, MCI vs. NC, MCI-C vs. MCI-NC, respectively, which are higher than the other methods. For the other three evaluation indicators of SEN, SPE and AUC, our method still outperforms the comparison methods on the different classification tasks. On ADNI-II, our method also achieves the promising classification performance in the experiment. The proposed method achieves the highest ACC and SPE values on the task of AD vs. NC, and it outperforms the other methods under the four used evaluation indicators on the task of MCI vs. NC.

B. Ablation Study

Ablation experiments are conducted on the proposed model to verify the role of each module of the proposed method. The accuracies under different ablation conditions on ADNI-I and ADNI-II are respectively shown in Table V and Table VI. It can be seen from the tables that among GCCA constraint, self-expression constraint and block-diagonal constraint, GCCA constraint has the greatest impact on classification accuracy. Common representation space is vital for comparability and complementarity of multi-modal data. In addition, both the self-expression constraint and the block-diagonal constraint have improved the diagnostic performance in AD diagnosis. For the choice of encoder, nonlinear encoder has more advantages in model performance than linear encoder.

In addition, we explore the influence of the block-diagonal constraint on self-expression coefficients. Specifically, we take advantage of samples in the same order to compare the coefficients with or without the block-diagonal constraint. As shown

TABLE V
THE CLASSIFICATION RESULTS UNDER DIFFERENT ABLATION CONDITIONS ON ADNI-I

Method	AD vs. NC				MCI vs. NC				MCI-C vs. MCI-NC			
	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
Without-GB ¹	87.78	93.23	84.78	91.97	75.08	84.75	58.35	79.23	69.66	68.17	70.69	70.58
Without-SB ²	88.85	93.14	85.67	92.86	78.99	86.50	64.05	79.87	72.68	69.25	75.13	71.31
Without-G ³	89.05	94.90	83.99	93.74	77.02	85.01	63.62	79.69	70.67	67.67	73.36	71.49
Without-B ⁴	92.36	94.14	91.29	95.53	81.55	92.55	60.12	80.06	80.16	76.45	83.57	75.72
Linear encoder ⁵	91.52	95.19	89.81	96.96	80.44	88.12	67.97	79.14	77.78	72.08	82.13	73.38
OUR	96.75	98.57	95.50	98.38	86.56	93.34	72.38	83.73	84.38	80.54	86.90	77.31

¹ Without-GB = remove GCCA and block-diagonal constraint

² Without-SB = remove self-expression and block-diagonal constraint

³ Without-G = remove GCCA

⁴ Without-B = remove block-diagonal constraint

⁵ Linear encoder = replace nonlinear encoder with linear encoder

TABLE VI
THE CLASSIFICATION RESULTS UNDER DIFFERENT ABLATION CONDITIONS IN ADNI-II

Method	AD vs. NC				MCI vs. NC			
	ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
Without-GB ¹	84.08	89.38	79.81	93.20	76.07	86.01	45.08	79.86
Without-SB ²	85.58	91.67	80.95	95.89	82.22	93.95	56.31	90.79
Without-G ³	84.91	92.15	78.01	93.19	79.53	84.66	55.61	84.71
Without-B ⁴	88.28	92.06	83.33	95.01	84.01	95.52	58.64	91.08
Linear encoder ⁵	88.14	92.59	84.30	93.13	80.63	89.75	57.65	85.92
OUR	92.45	94.27	90.44	96.27	86.75	95.15	67.86	91.18

¹ Without-GB = remove GCCA and block-diagonal constraint

² Without-SB = remove self-expression and block-diagonal constraint

³ Without-G = remove GCCA

⁴ Without-B = remove block-diagonal constraint

⁵ Linear encoder = replace nonlinear encoder with linear encoder

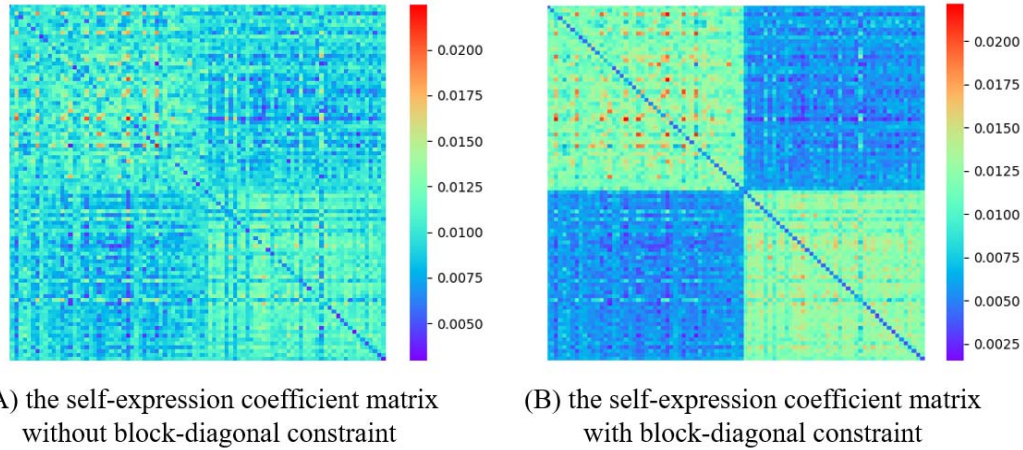


Fig. 3. Influence of block-diagonal constraint on self-expression coefficient matrix.

in Fig. 3(A), by the model without block-diagonal constraint, the diagonal block structure of the self-expression coefficients exist, but it is not obvious. It indicates that the representation learned is not enough to clearly distinguish the class of samples. Therefore, it is critical to embed structural information to enhance diagonal block structure in the self-expression coefficient matrix, as shown in Fig. 3(B). The block-diagonal constraint promotes discriminative representation to obtain better classification performance.

C. Convergence Analysis

The convergence curve of the objective function with all tasks in the experiment is shown in Fig. 4, where Fig. 4(A),

Fig. 4(B), Fig. 4(C) respectively show the curves of AD vs. NC, MCI vs. NC, and MCI-C vs. MCI-NC on ADNI-I, and Fig. 4(D), Fig. 4(E) respectively show the curves of AD vs. NC, MCI vs. NC on ADNI-II. In this figure, the convergence curve on the two datasets decrease sharply in the previous rounds of iterations, and both began to converge after the 50th iterations. The convergence curve tends to be stable, which shows that our objective function is convergent.

D. Parameters Analysis

In our proposed method, the objective function introduces three regularization parameters, i.e. λ_1 , λ_2 , λ_3 . These

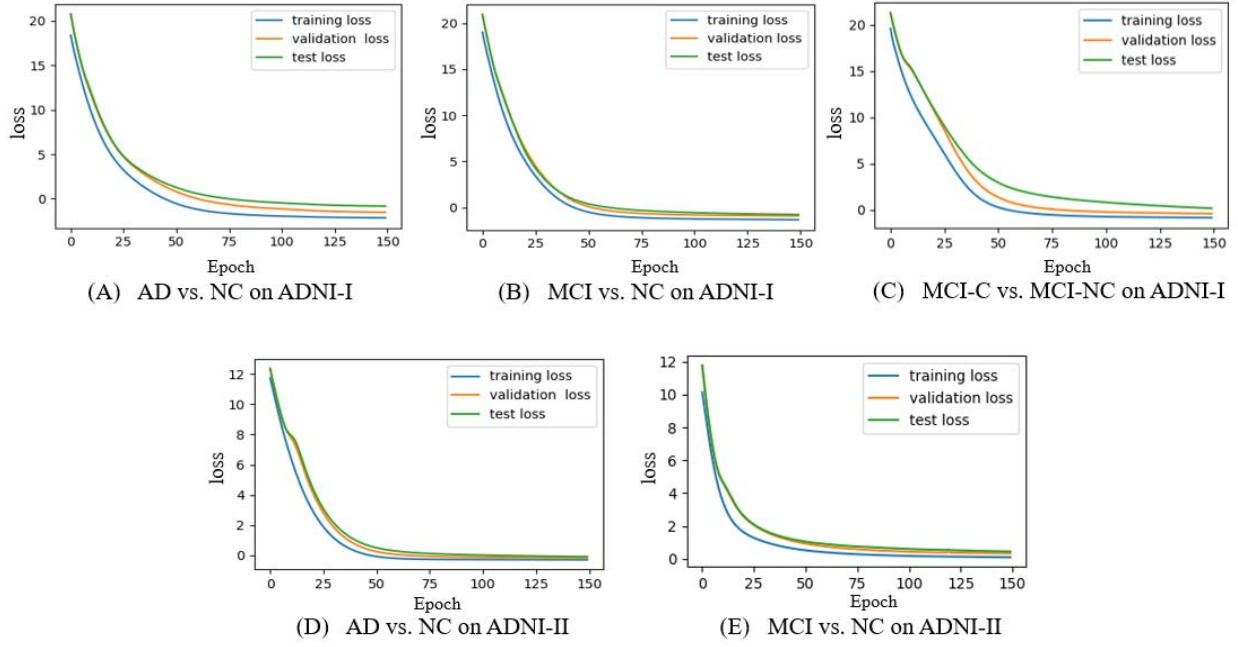


Fig. 4. The convergence curves of different tasks on ADNI-I and ADNI-II.

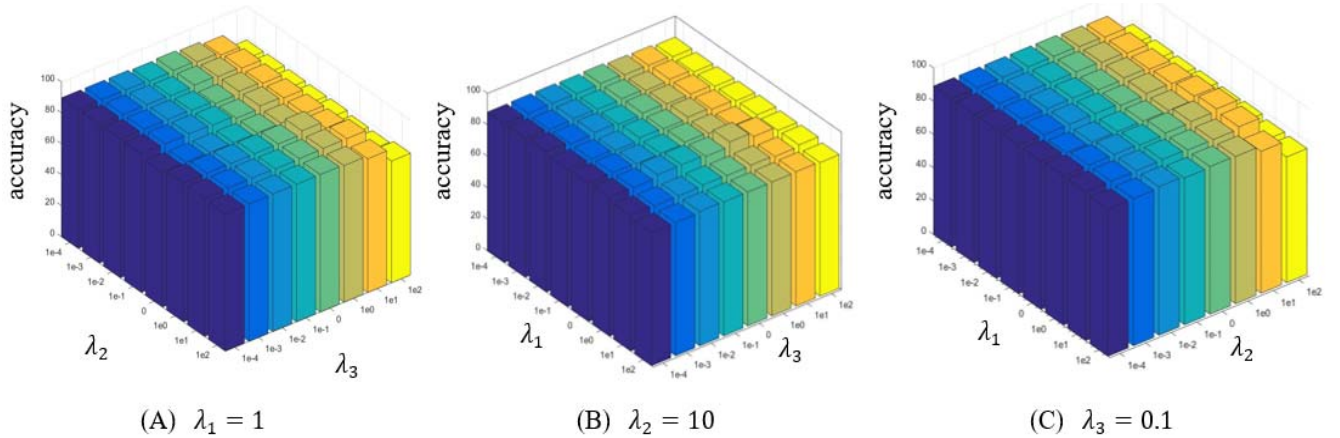


Fig. 5. Performance of classification accuracy under different parameters.

regularization parameters are used to balance the proportions of self-expression constraint, block-diagonal constraint and GCCA constraint. We explore the classification accuracy under different parameters on ADNI-II with task AD vs. NC. The regularization parameters λ_1 , λ_2 , λ_3 are set in the range of $\{1e-4, 1e-3, 1e-2, 1e-1, 0, 1e0, 1e1, 1e2\}$. Grid search is adopted to determine the optimal parameters through the validation set accuracy under the combinations of above three parameters, and the optimal values are $\lambda_1 = 1$, $\lambda_2 = 10$, $\lambda_3 = 0.1$. For the convenience of display, we fix one parameter as the optimal value in turn and control the other two parameters in pairs to show the accuracy changes under various combinations, and the results are shown in Fig. 5. Experiments show that our proposed method is not sensitive to parameter.

E. Model Distillation and Biomarker Analysis

We propose LMKD module to seek the significant ROIs in AD diagnosis. Specifically, for a certain modality, we train

the corresponding LMKD and utilize SVM to classify the new generated representation under LMKD. We multiply the weight of the feature mapping matrix in LMKD and the feature score in SVM as the ROI impact score. We verify the module on ADNI-II with task of AD vs. NC, and evaluate the effectiveness from two aspects: classification accuracy and ROIs analysis.

For accuracy, we compare the performance of LMKD with the first layer weight (FLW) [12] and DMDR, under different conditions of single modality and multi-modality. The results are shown in Table VII. In the table, we can see that LKMD performs better in both single-modality and multi-modality compared to FLW that only uses the first-level weight as the projection matrix. It shows that global weights are more effective than partial weights. Compared with DMDR, the classification performance of LKMD has not dropped sharply. It shows that LKMD effectively retains a certain multi-modal potential structure in DMDR.

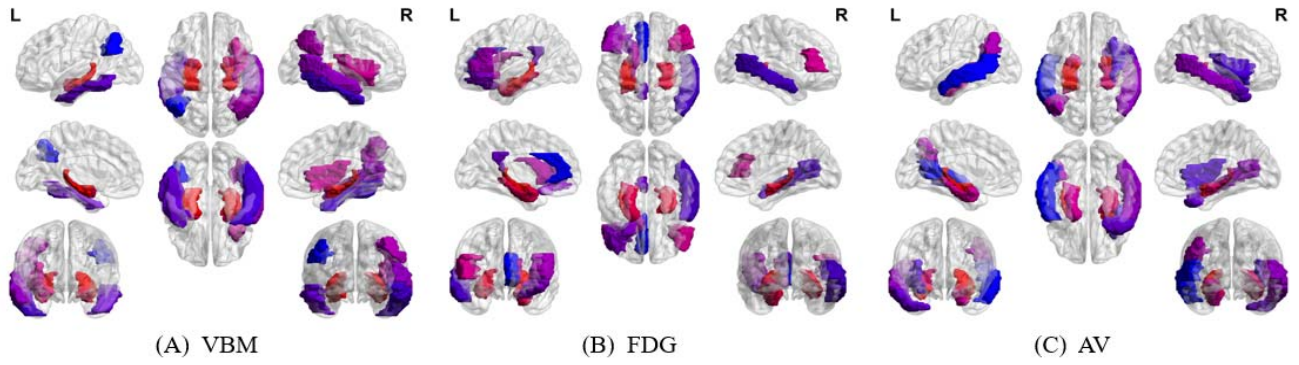


Fig. 6. The Top-10 influential ROIs in AD diagnosis.

TABLE VII

THE CLASSIFICATION RESULTS OF OUR METHOD AND OTHER COMPARED METHODS ON ADNI-II

Method	Modality	ACC	SEN	SPE	AUC
FLW	VBM	87.07	91.45	81.64	92.59
	FDG	80.83	87.81	71.35	85.21
	AV	85.45	91.54	78.11	91.51
	CONCAT	88.13	91.18	85.10	93.99
LKMD	VBM	88.10	92.25	83.53	92.33
	FDG	81.35	87.29	74.51	86.02
	AV	85.68	89.73	81.07	91.59
	CONCAT	89.45	90.09	88.16	94.48
DMDR	VBM	89.21	92.86	85.63	93.35
	FDG	81.97	86.08	78.07	86.52
	AV	87.60	93.12	81.17	90.23
	CONCAT	92.45	94.27	90.44	96.27

For ROIs analysis, we calculate the ROI impact score according to the method described in the previous paragraph. We select the top-10 ROIs with the highest scores in the three modalities and display them in Fig. 6. The redder ROI is, the easier it is to distinguish AD/NC. All three modalities show that the hippocampus is the most influential ROI in our model. Hippocampus has been identified as the most recognized one among AD core markers, which is related to memory system [46]. Among the three modalities, the ROIs selected by VBM and AV are quite similar, including amygdala, insula, angular and temporal_mid. Temporal_inf is observed in VBM, while the temporal_pole_mid in AV. In addition, the ROIs selected by FDG is mostly present in the frontal lobe (inferior frontal, rectus). The experimental results are roughly consistent with previous Alzheimer's disease pathological studies. In the early stage of Alzheimer's disease, the medial temporal lobe structures are mainly affected (hippocampus, amygdala, etc.) with higher atrophy compared to normal aging [47], which leads to some obstacles in recent episodic memory. As the disease progressed, the abnormal brain regions gradually spread, with significant changes in the frontal and temporoparietal association cortices, cingulate gyrus, insula and precuneus, as well as in the thalamus and basal ganglia [48]. Hypometabolism in the anterior cingulate cortex and hypoperfusion in the left middle posterior frontal gyrus near the infra frontal junction can cause attention/execution deficits. Visuospatial and perceptual symptoms usually appear with episodic memory degradation and attention deficit. With

the alterations in the inferior parietal lobules extending to the intraparietal sulcus and in the precuneus, visuospatial information processing ability and spatial guidance ability degrade. Anosognosia may be related to the anterior medial frontal cortex/pregenual anterior cingulate together with the posterior cingulate cortex and temporoparietal junction area [49].

VI. CONCLUSION

In this paper, a deep multi-modal discriminative and interpretability network is proposed for multi-modal AD diagnosis. In our method, each modality is reconstructed into a hierarchical representation through MLP, and discriminative self-expression learning is adopted to achieve inter-modal and intra-modal sample alignment by embedding the class structure information into the self-expression coefficients. Further, with the GCCA constraint, different modalities are finally projected into a common discriminative space, in which the subspaces are used for AD diagnosis. In terms of model interpretability, a linear multi-modal distillation module is proposed to explore the influence of brain regions in AD diagnosis. It reversely extracts feature weights by using linear mapping to learn the coordinated representation. The proposed method and module are verified on the two datasets ADNI-I and ADNI-II, and the results show the effectiveness of the proposed network.

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